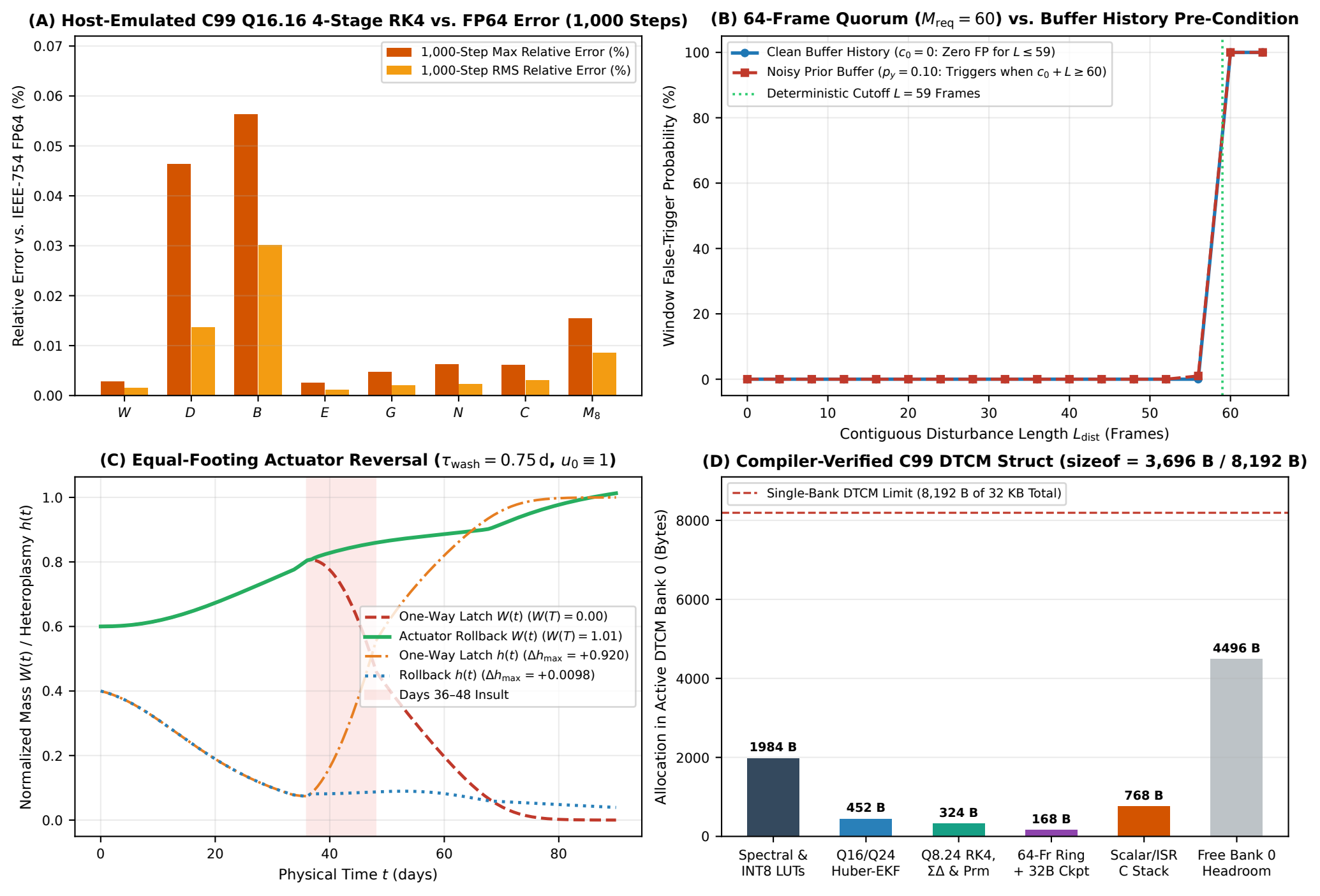
(A) Host-Emulated C99 Q16.16 4-Stage RK4 vs. FP64 Error (1,000 Steps)
1,000-Step Max Relative Error (%)
1,000-Step RMS Relative Error (%)
Relative Error vs. IEEE-754 FP64 (%)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
W
D
B
E
G
N
C
M8
(B) 64-Frame Quorum ($M_{req} = 60$) vs. Buffer History Pre-Condition
Clean Buffer History ($c_0 = 0$: Zero FP for $L \le 59$)
Noisy Prior Buffer ($p_y = 0.10$: Triggers when $c_0 + L \ge 60$)
Deterministic Cutoff $L = 59$ Frames
Window False-Trigger Probability (%)
0
20
40
60
80
100
0
10
20
30
40
50
60
Contiguous Disturbance Length $L_{dist}$ (Frames)
(C) Equal-Footing Actuator Reversal ($\tau_{wash} = 0.75$ d, $u_0 \equiv 1$)
One-Way Latch $W(t)$ ($W(T) = 0.00$)
Actuator Rollback $W(t)$ ($W(T) = 1.01$)
One-Way Latch $h(t)$ ($\Delta h_{max} = +0.920$)
Rollback $h(t)$ ($\Delta h_{max} = +0.0098$)
Days 36–48 Insult
Normalized Mass $W(t)$ / Heteroplasmy $h(t)$
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
Physical Time $t$ (days)
(D) Compiler-Verified C99 DTCM Struct (sizeof = 3,696 B / 8,192 B)
Single-Bank DTCM Limit (8,192 B of 32 KB Total)
Allocation in Active DTCM Bank 0 (Bytes)
0
2000
4000
6000
8000
1984 B
452 B
324 B
168 B
768 B
4496 B
Spectral & INT8 LUTs
Q16/Q24 Huber-EKF
Q8.24 RK4, ΣΔ & Prm
64-Fr Ring + 32B Ckpt
Scalar/ISR C Stack
Free Bank 0 Headroom